

CAPE ROMANZOF, AK, USA

WMO: 702120

Lat: 61.783N

Lon: 166.033W

Elev: 479

StdP: 14.44

Time Zone: -9.00 (NAK)

Period: 94-19

WBAN: 26646

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-13.4	-10.0	-19.7	1.9	-11.8	-16.7	2.2	-8.0	56.7	6.4	52.0	2.2	21.2	10	0.999

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	7.3	65.3	56.3	62.0	54.9	58.9	53.4	57.9	63.4	55.8	60.4	54.2	58.0	11.0	40

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
55.3	66.2	59.1	53.6	62.3	57.4	52.1	58.9	55.9	25.2	63.4	23.9	60.6	22.9	57.9	68.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 46.6	40.8	36.1	-12.6	70.5	5.5	4.4	-16.5	73.7	-19.7	76.3	-22.8	78.8	-26.8	82.0		
(5)			-13.0	59.9	5.4	3.0	-16.8	62.1	-20.0	63.9	-23.0	65.6	-26.9	67.8		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	31.4	13.0	16.0	15.4	25.0	36.9	46.3	51.0	50.7	45.4	34.6	24.0	18.1
	DBStd	16.79	14.56	13.37	11.96	10.14	7.81	6.49	5.12	4.93	5.18	6.42	8.86	11.22
	HDD50	6974	1147	953	1072	750	412	148	45	47	153	476	781	990
	HDD65	12250	1612	1373	1537	1200	871	562	435	444	589	941	1231	1455
	CDD50	201	0	0	0	0	6	36	76	69	14	0	0	0
	CDD65	3	0	0	0	0	0	0	2	1	0	0	0	0
	CDH74	6	0	0	0	0	0	0	6	0	0	0	0	0
	CDH80	0	0	0	0	0	0	0	0	0	0	0	0	0
Wind	WSAvg	16.4	20.7	20.4	19.0	17.3	14.1	11.6	11.3	12.3	15.0	16.5	18.8	20.0
Precipitation	PrecAvg	28.1	1.9	1.9	1.3	1.2	1.5	2.1	3.0	3.9	3.8	2.8	2.6	2.1
	PrecMax	38.3	4.3	4.6	4.2	2.4	3.2	3.2	4.6	5.8	6.7	5.6	5.1	4.4
	PrecMin	22.6	0.4	0.6	0.2	0.3	0.2	0.9	1.8	1.3	0.6	0.8	0.5	0.5
	PrecStd	3.7	0.9	1.0	0.9	0.5	0.7	0.6	0.8	1.0	1.1	1.0	1.2	1.1
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	42.8	39.0	38.4	47.3	62.8	68.5	73.2	69.4	61.3	50.9	44.5	40.8
		MCWB	35.4	34.3	33.6	39.7	50.6	55.7	61.4	59.1	52.6	45.4	41.9	36.0
	2%	DB	38.2	35.7	35.6	43.3	55.8	62.7	65.7	64.2	56.4	47.2	40.2	37.0
		MCWB	33.9	32.6	32.2	38.0	46.8	53.4	56.3	57.2	51.5	44.8	38.1	34.4
	5%	DB	34.6	33.5	33.4	40.6	51.3	59.0	61.8	60.8	53.9	45.3	37.6	34.4
		MCWB	32.4	31.7	31.4	36.3	44.4	51.8	55.4	55.9	50.6	43.5	36.2	32.6
	10%	DB	32.2	32.1	31.3	37.9	47.8	55.7	58.5	57.4	51.7	43.1	35.7	32.4
		MCWB	30.7	30.7	29.7	34.5	42.5	50.3	53.6	54.3	49.1	41.3	34.6	31.1
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	37.8	35.6	35.3	40.3	51.8	57.2	61.2	61.3	54.6	47.3	42.3	38.0
		MCDB	41.2	37.4	37.3	45.9	59.8	65.4	72.4	66.1	58.3	48.9	44.0	38.9
	2%	WB	34.9	33.4	33.1	38.5	47.9	54.7	58.0	58.3	52.6	45.3	38.6	34.9
		MCDB	37.0	34.5	34.5	42.8	54.1	60.8	64.0	62.8	55.2	46.6	39.8	36.4
	5%	WB	32.6	32.2	31.7	36.6	45.4	52.7	55.7	56.3	50.9	43.6	36.5	33.0
		MCDB	34.2	33.2	33.2	40.1	50.5	57.9	60.6	60.1	53.4	45.3	37.5	34.2
	10%	WB	30.6	30.9	29.9	34.9	42.9	50.7	54.1	54.4	49.6	41.7	34.5	31.4
		MCDB	32.3	32.0	31.1	37.7	47.3	55.4	58.1	57.2	51.6	43.0	35.5	32.6
Mean Daily Temperature Range	5% DB	MDBR	9.6	9.9	9.7	8.9	8.5	8.9	7.3	5.9	5.5	5.5	7.3	8.9
		MCDBR	11.2	9.2	9.7	9.7	13.5	13.9	13.0	10.5	7.9	5.8	6.7	8.8
		MCWBR	9.2	8.0	8.1	6.5	8.7	8.7	8.2	6.9	5.8	5.1	6.0	7.4
	5% WB	MCDBR	10.7	8.8	8.8	9.3	12.5	13.0	11.1	9.3	7.1	5.6	6.6	7.4
		MCWBR	9.3	8.1	7.8	6.6	8.5	8.7	7.5	6.6	5.6	5.5	6.3	6.7
Clear-Sky Solar Irradiance	taub	0.212	0.277	0.282	0.345	0.364	0.364	0.370	0.383	0.335	0.300	0.250	0.203	
	taud	2.035	2.122	2.200	2.061	2.215	2.414	2.441	2.370	2.476	2.358	2.032	1.879	
	Ebn at Noon	157	215	264	264	270	273	268	253	247	210	129	89	
	Edh at Noon	11	22	29	42	39	33	31	31	23	18	12	8	
All-Sky Solar Radiation	RadAvg	96	329	878	1418	1600	1638	1328	1048	679	330	101	43	
	RadStd	14	42	103	126	190	150	95	135	60	26	16	5	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (6 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page